

Finite-Component Structure of Mixed-Unitary Times

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Status

This packet records a candidate partial result, likely valid, toward the open remark following Theorem 5.4 of Bhat and Devendra [1]. The source asks whether the first mixed-unitary re-entry time must already be the beginning of the eventual mixed-unitary ray. We do not decide that existence question. Instead, we prove that the full set of mixed-unitary times always has only finitely many connected components.

Source Passage

a real number. This allows us to define a real valued function g on $[0, \infty)$ by $g(t) := \langle \Gamma, e^{t\mathcal{L}} \rangle$, $t \in [0, \infty)$. Then g is a differentiable function with $g(0) = 0$. Furthermore,

$$g'(0) = \lim_{t \rightarrow 0} \frac{\langle \Gamma, e^{t\mathcal{L}} \rangle - \langle \Gamma, \text{id} \rangle}{t} = \langle \Gamma, \mathcal{L} \rangle < 0.$$

Therefore, there exists a $s_0 > 0$ such that $g(t) < 0$ for all $t \in (0, s_0)$. i.e., $\langle \Gamma, \Phi_t \rangle < 0$ for all $0 < t < s_0$. Now, by Lemma 2.6 we get that Φ_t is not mixed unitary for all $t \in (0, s_0)$.

Let $t_0 := \max\{t : \Phi_s \text{ is not mixed unitary}, \forall s \in (0, t)\}$. We claim that Φ_{t_0} is a mixed unitary. In deed this clear from openness of N , as otherwise $t_0 \in N$ and this would mean $[t_0, t_0 + \epsilon) \subseteq N$ for some $\epsilon > 0$. Now, define

$$t_1 := \min\{t : \Phi_s \text{ is mixed unitary}, \forall s \in [t, \infty)\}.$$

By Theorem 4.12 t_1 is finite. Clearly $t_0 \leq t_1$. □

We wish to remark that in the previous theorem, we do not know whether one can take $t_1 = t_0$ or not. Considering the general features of numerous situations of eventual positivity (See for example [21]) we would guess that there could be examples for which $t_0 < t_1$ and the set of points t in $[t_0, t_1]$ for which Φ_t is a mixed unitary channel is a discrete set. We are not able to confirm this in any of the following examples.

We now give an example of a dynamical semigroup of unital quantum channels, which is not mixed unitary for some $t > 0$. For any $B \in \mathbb{M}_n$ let $u(B)$ denote the set of all singular

Question and Answered Part

Let $(\Phi_t)_{t \geq 0}$, with $\Phi_t = e^{t\mathcal{L}}$, be a quantum dynamical semigroup of unital channels on $\mathbb{M}_d$. The source proves that either every Φ_t is mixed unitary, or there are $0 < t_0 \leq t_1 < \infty$ such that

$$\Phi_t \notin \text{MU}(d) \quad (0 < t < t_0), \quad \Phi_{t_0} \in \text{MU}(d), \quad \Phi_t \in \text{MU}(d) \quad (t \geq t_1). \quad (1)$$

It asks whether $t_0 = t_1$ must hold. If not, the authors suggest that the mixed-unitary times in $[t_0, t_1]$ might form a discrete set.

We prove that arbitrary infinite oscillation cannot occur: the complete time set is a finite union of intervals and points. In particular, a discrete intermediate set must be finite.

Proof Intuition

The set of mixed-unitary channels in fixed dimension is semialgebraic. Indeed, the unitary group is cut out by polynomial equations in real and imaginary parts, unitary conjugation depends polynomially on those coordinates, and Carathéodory's theorem gives a uniform finite bound on the number of terms in a convex combination. Tarski–Seidenberg then removes the auxiliary unitaries and convex coefficients.

A quantifier-free semialgebraic description uses the signs of only finitely many polynomials. On the analytic curve $t \mapsto e^{t\mathcal{L}}$, those polynomials become finitely many real-analytic functions. A nonzero real-analytic function has only finitely many zeros on a compact interval. Consequently the membership pattern can change only finitely many times.

Semialgebraicity

We regard the real vector space of Hermiticity-preserving linear maps $\mathbb{M}_d \rightarrow \mathbb{M}_d$ as $\mathbb{R}^N$, after fixing real coordinates.

Lemma 1. *For every d , the set $\text{MU}(d)$ of mixed-unitary channels on $\mathbb{M}_d$ is a compact semialgebraic subset of $\mathbb{R}^N$.*

Proof. Write a complex $d \times d$ matrix in real and imaginary coordinates. The unitary group $U(d)$ is semialgebraic because $U^*U = I$ is a finite system of real polynomial equations. In the fixed coordinates on the space of linear maps, the entries of

$$\text{Ad}_U : X \mapsto UXU^*$$

are real polynomials in the entries of U .

By Carathéodory's theorem, every point in the convex hull of a subset of $\mathbb{R}^N$ is a convex combination of at most $N + 1$ points. Hence $\Phi \in \text{MU}(d)$ if and only if there exist

$$U_0, \dots, U_N \in U(d), \quad \lambda_0, \dots, \lambda_N \geq 0, \quad \sum_{j=0}^N \lambda_j = 1,$$

such that

$$\Phi = \sum_{j=0}^N \lambda_j \text{Ad}_{U_j}. \quad (2)$$

All conditions in (2) are polynomial equalities or inequalities in finitely many real variables. Thus the set of tuples $(\Phi, \lambda_0, \dots, \lambda_N, U_0, \dots, U_N)$ satisfying them is semialgebraic. Its projection to the Φ -coordinate is semialgebraic by the Tarski–Seidenberg theorem [2, Chapter 2].

Finally, $U(d)$ is compact, and the convex hull of a compact subset of a finite-dimensional space is compact. Therefore $\text{MU}(d)$ is compact. $\square$

Finite-Component Theorem

Theorem 1. *Let $(\Phi_t)_{t \geq 0}$, $\Phi_t = e^{t\mathcal{L}}$, be a quantum dynamical semigroup of unital channels on $\mathbb{M}_d$, and define*

$$S_{\mathcal{L}} = \{t \geq 0 : \Phi_t \in \text{MU}(d)\}. \quad (3)$$

Then:

1. $S_{\mathcal{L}}$ is a closed additive subsemigroup of $[0, \infty)$ containing 0;
2. $S_{\mathcal{L}}$ has finitely many connected components;
3. more explicitly, $S_{\mathcal{L}}$ is a finite union of isolated points and closed intervals, one interval being a ray $[\tau, \infty)$.

Proof. The identity channel Φ_0 is unitary, so $0 \in S_{\mathcal{L}}$. The set is closed because $t \mapsto \Phi_t$ is continuous and $\text{MU}(d)$ is closed. If $s, t \in S_{\mathcal{L}}$, then $\Phi_{s+t} = \Phi_s \circ \Phi_t$ is mixed unitary: composing two convex combinations of unitary conjugations gives another such combination. Therefore $s + t \in S_{\mathcal{L}}$.

By eventual mixed unitarity, Theorem 4.12 of [1], there exists $T < \infty$ such that

$$[T, \infty) \subseteq S_{\mathcal{L}}. \quad (4)$$

It remains to prove that $S_{\mathcal{L}} \cap [0, T]$ has finitely many components.

By Lemma 1 and quantifier elimination for semialgebraic sets, membership in $\text{MU}(d)$ has a quantifier-free description involving the signs of finitely many real polynomials

$$P_1, \dots, P_m$$

in the coordinates of a linear map. Put

$$f_j(t) = P_j(\Phi_t) = P_j(e^{t\mathcal{L}}).$$

Every coordinate of the finite-dimensional matrix exponential $e^{t\mathcal{L}}$ is real analytic in t . Hence every f_j is real analytic on $\mathbb{R}$.

If f_j is not identically zero, it has only finitely many zeros in $[0, T]$. Indeed, infinitely many zeros in that compact interval would have an accumulation point, forcing a real-analytic function to vanish identically. Functions that are identically zero impose a constant sign condition. Let Z be the union of the zero sets in $[0, T]$ of all nonzero f_j . Then Z is finite.

On every component of $[0, T] \setminus Z$, the sign of every f_j is constant. The quantifier-free membership formula therefore has a constant truth value on that component. It follows that $S_{\mathcal{L}} \cap [0, T]$ is a finite union of intervals and points. Because $S_{\mathcal{L}}$ is closed, the intervals may be taken closed after merging adjacent pieces. Combining them with the ray (4) proves the result. $\square$

Corollary 1 (The discrete scenario is finite). *Assume the nontrivial alternative (1) from the source paper. Then the mixed-unitary times in $[t_0, t_1]$ form a finite union of closed intervals and isolated points. In particular, if that set is discrete, it is finite.*

Proof. Apply Theorem 1. A closed discrete subset of a compact interval is finite. $\square$

Corollary 2 (An interval forces a ray). *If $0 < a < b$ and $[a, b] \subseteq S_{\mathcal{L}}$, then, for*

$$n_0 = \left\lceil \frac{a}{b-a} \right\rceil,$$

one has

$$[n_0 a, \infty) \subseteq S_{\mathcal{L}}. \quad (5)$$

In particular, an interval $[0, b] \subseteq S_{\mathcal{L}}$ forces $S_{\mathcal{L}} = [0, \infty)$.

Proof. Additivity gives

$$m[a, b] = [ma, mb] \subseteq S_{\mathcal{L}} \quad (m \in \mathbb{N}).$$

For every $m \geq n_0$,

$$mb \geq (m+1)a \iff m(b-a) \geq a.$$

Thus the intervals $[ma, mb]$, $m \geq n_0$, overlap consecutively, and their union is $[n_0 a, \infty)$. If $[0, b] \subseteq S_{\mathcal{L}}$, every $t \geq 0$ is a sum of finitely many elements of that interval. $\square$

Limitations

Theorem 1 rules out a Cantor-like time set and rules out infinitely many isolated re-entries before eventual mixed unitarity. It does not prove $t_0 = t_1$. For example, at the purely additive level, a set such as

$$\{0, a\} \cup [2a, \infty)$$

is a closed additive subsemigroup with a genuine gap. The remaining problem is whether the analytic curve generated by a Lindbladian can realize such a pattern inside the particular convex body $\text{MU}(d)$.

Novelty Check and Human Review

The run indexes and bounded external searches on 23 July 2026 were checked using the arXiv id, exact title, mixed-unitary time sets, semialgebraicity, finite unions of intervals, and semigroup formulations. They found the source preprint and adjacent mixed-unitarity work, but no statement of Theorem 1. This is evidence, not a guarantee, of novelty.

Human review should focus on Lemma 1. Carathéodory gives a uniform finite number of auxiliary unitary matrices, so Tarski–Seidenberg applies in a fixed finite-dimensional real space. After that step, the one-variable analytic argument is direct. The packet should remain a partial result until the existence or impossibility of $t_0 < t_1$ is resolved.

References

- [1] B. V. Rajarama Bhat and R. Devendra, *On regions of mixed unitarity for semigroups of unital quantum channels*, arXiv:2512.23598, version of 9 February 2026. arXiv:2512.23598.
- [2] S. Basu, R. Pollack, and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second edition, Algorithms and Computation in Mathematics 10, Springer, 2006.